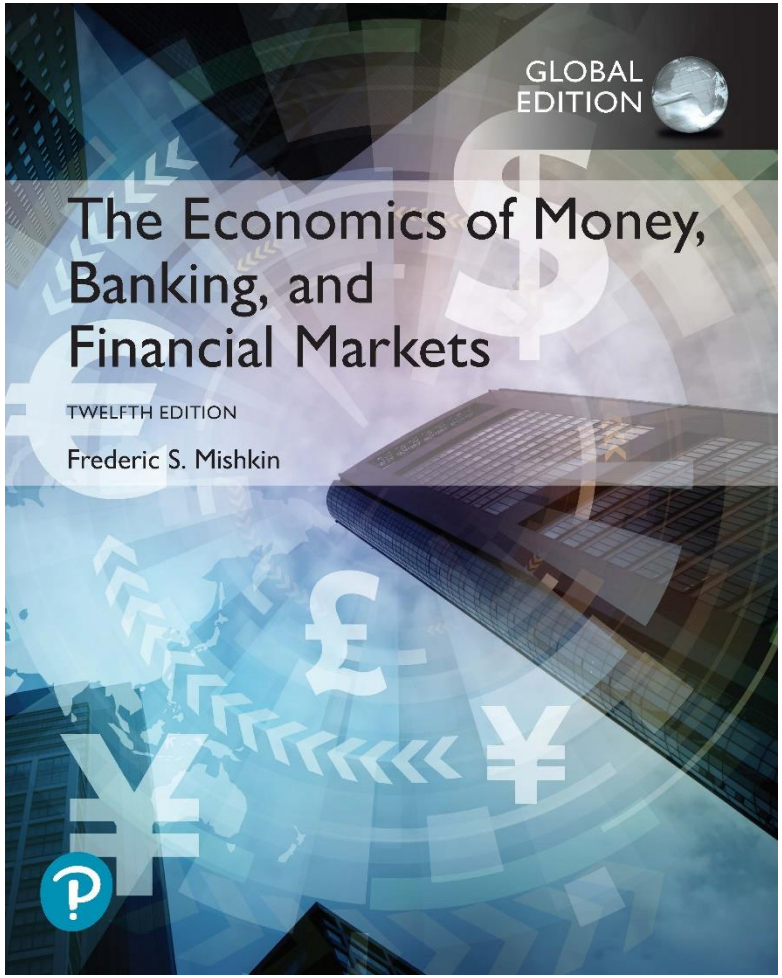
The Economics of Money, Banking, and Financial Markets

Twelfth Edition, Global Edition



Chapter 10

Economic Analysis of Financial Regulation

Preview

- This chapter develops an economic analysis of financial regulation.

Learning Objectives

- Identify the reasons for and forms of a government safety net in financial markets.
- List and summarize the types of financial regulation and how each reduces asymmetric information problems.

Asymmetric Information as a Rationale for Financial Regulation

- Bank panics and the need for deposit insurance:
  - FDIC: short circuits bank failures and contagion effect
  - Payoff method
  - Purchase and assumption method (typically more costly for the FDIC)
- Other form of government safety net:
  - Lending from the central bank to troubled institutions (lender of last resort)

The Spread of Government Deposit Insurance Throughout the World: Is This a Good Thing?

- Has government deposit insurance helped improve the performance of the financial system and prevent banking crises? The answer seems to be “no.” Research at the World Bank seems to answer “no,” since on average, the adoption of explicit government deposit insurance is associated with less banking sector stability and a higher incidence of banking crises. Furthermore, on average, deposit insurance seems to retard financial development.

Drawbacks of the Government Safety Net

- **Moral Hazard**
  - Depositors do not impose discipline of marketplace
  - Financial institutions have an incentive to take on greater risk
- **Adverse Selection**
  - Risk-lovers find banking attractive
  - Depositors have little reason to monitor financial institutions

“Too Big to Fail”

- Government provides guarantees of repayment to large uninsured creditors of the largest financial institutions even when they are not entitled to this guarantee.
- Uses the purchase and assumption method
- Increases moral hazard incentives for big banks

Financial Consolidation and the Government Safety Net

- Larger and more complex financial organizations challenge regulation:
  - Increased “too big to fail” problem
  - Extends safety net to new activities, increasing incentives for risk-taking in these areas (as has occurred during the global financial crisis

---- “too complex to fail” , e.g., AIG

Types of Financial Regulation: Restrictions on Asset Holdings

- Attempts to restrict financial institutions from too much risk taking:
  - Bank regulations
    - Promote diversification
    - Prohibit holdings of common stock
  - Capital requirements
    - Minimum leverage ratio (for banks)
    - Basel Accord: risk-based capital requirements
    - Regulatory arbitrage

Types of Financial Regulation: Capital Requirements

- Government-imposed capital requirements are another way of minimizing moral hazard at financial institutions
- There are two forms:
  1. Based on the leverage ratio, the amount of capital divided by the bank's total assets: to be classified as well capitalized, a bank's leverage ratio must exceed 5%; a lower leverage ratio, especially one below 3%, triggers increased regulatory restrictions on the bank.
  2. Risk-based capital requirements

  Regulatory arbitrage problem

Where Is the Basel Accord Heading After the Global Financial Crisis? (1 of 3)

- Starting in June 1999, the Basel Committee on Banking Supervision released several proposals to reform the original 1988 Basel Accord. These efforts have culminated in what bank supervisors refer to as Basel 2, which is based on three pillars.

Where Is the Basel Accord Heading After the Global Financial Crisis? (2 of 3)

- Pillar 1 links capital requirements for large, internationally active banks more closely to actual risk of three types: market risk, credit risk, and operational risk.
- Pillar 2 focuses on strengthening the supervisory process, particularly in assessing the quality of risk management in banking institutions and evaluating whether these institutions have adequate procedures in place for determining how much capital they need.

Where Is the Basel Accord Heading After the Global Financial Crisis? (3 of 3)

- Pillar 3 focuses on improving market discipline through increased disclosure of details about a bank's credit exposures, its amount of reserves and capital, the officials who control the bank, and the effectiveness of its internal rating system.

Basel Accord （巴塞尔协议）

- **Basel I**

Capital requirement primarily focuses on credit risk and appropriate risk-weighting of assets.

Assets of banks were classified and grouped in different categories according to credit risk, carrying risk weights of 0% (e.g., cash), 20% (e.g., MBS with AAA rating), 50% (e.g., residential mortgages), 100% (e.g., most corporate debt).

- **Basel II**

Capital requirement is calculated for three major components of risk that a bank faces: credit risk, operational risk, and market risk.

Ensuring that credit risk, operational risk and market risk are quantified based on data and formal techniques

- **Basel III**

Strengthening the requirements from the Basel II standard on bank's minimum capital ratios, it introduces requirements on liquid asset holdings and funding stability.

(minimum capital ratio, liquidity coverage ratio, stable funding ratio)

Financial Supervision: Prompt Corrective Action

- If the amount of a financial institution's capital falls to low levels, serious problems result.
- To prevent this, the Federal Deposit Insurance Corporation Improvement Act of 1991 adopted prompt corrective action provisions that require the FDIC to intervene earlier and more vigorously when a bank gets into trouble.

possible negative externality across banks

Financial Supervision: Chartering and Examination

- **Chartering** (screening of proposals to open new financial institutions) to prevent adverse selection
- **Examinations** (scheduled and unscheduled) to monitor capital requirements and restrictions on asset holding to prevent moral hazard
  - Capital adequacy
  - Asset quality
  - Management
  - Earnings
  - Liquidity
  - Sensitivity to market risk
- Filing periodic “call reports”

Assessment of Risk Management

- Greater emphasis on evaluating soundness of management processes for controlling risk
- Trading Activities Manual of 1994 for risk management rating based on:
  - Quality of oversight provided
  - Adequacy of policies and limits for all risky activities
  - Quality of the risk measurement and monitoring systems
  - Adequacy of internal controls
- Interest-rate risk limits:
  - Internal policies and procedures
  - Internal management and monitoring
  - Implementation of stress testing and value-at risk (VaR)

Disclosure Requirements

- Requirements to adhere to standard accounting principles and to disclose wide range of information
- The Basel 2 accord and the SEC put a particular emphasis on disclosure requirements
- The Sarbanes-Oxley Act of 2002 established the Public Company Accounting Oversight Board
- Mark-to-market (fair-value) accounting

Disclosure Requirements

Mark-to-market accounting vs. historical cost accounting

Cifuentes, Ferrucci and Shin (JEEA, 2015)

- Disadvantage of mark-to-market accounting: contagious failures
- Advantage of mark-to-market accounting: transparent

When the market's demand for illiquid assets is less than perfectly elastic, sales by distressed institutions depress the market prices of such assets. Marking to market of the asset book can induce a further round of endogenously generated sales of assets, depressing prices further and inducing further sales. Contagious failures can result from small shocks.

Bleck and Liu (JAR, 2007)

- Disadvantage of historical cost accounting: veil and give managers a “call option”
- Advantage of historical cost accounting: no discretion in reporting

The paper shows that marking to market can provide investors with an early warning mechanism while historical cost gives management a “veil” under which they can potentially mask a firm's true economic performance. Historic-cost-accounting effectively gives a call option to the manager: reveal good and hide bad, leading to more frequent and more severe crashes in asset prices, and making the financial market more rather than less volatile, which runs counter to conventional wisdom.

Consumer Protection

- Consumer Protection Act of 1969 (Truth-in-lending Act)
- Fair Credit Billing Act of 1974
- Equal Credit Opportunity Act of 1974, extended in 1976
- Community Reinvestment Act
- The subprime mortgage crisis illustrated the need for greater consumer protection.

Examples:

-- **Predatory lending**

Any practice of a lender that imposes unfair and abusive loan terms on borrowers, including high interest rates, high fees, and terms that strip the borrower of equity.

-- **hidden or undisclosed fees**

30 Day Free Trial-3 Audiobooks in amazon, hard to cancel or refund

-- **insurance products**

claim is hard or with dispute

Restrictions on Competition

- Justified as increased competition can also increase moral hazard incentives to take on more risk.
  - Branching restrictions (eliminated in 1994)
  - Glass-Steagall Act (repeated in 1999)
- Disadvantages:
  - Higher consumer charges
  - Decreased efficiency

International Financial Regulation

- Particular problems in financial regulation occur when financial institutions operate in many countries and thus can shift their business readily from one country to another. Financial regulators closely examine the domestic operations of financial institutions in their own country, but they often do not have the knowledge or ability to keep a close watch on operations in other countries

Table 1 Major Financial Legislation in the United States (1 of 5)

Federal Reserve Act (1913)
- Created the Federal Reserve System

McFadden Act of 1927
- Effectively prohibited banks from branching across state lines
- Put national and state banks on equal footing regarding branching

Banking Acts of 1933 (Glass-Steagall) and 1935
- Created the FDIC
- Separated commercial banking from the securities industry
- Prohibited interest on checkable deposits and restricted such deposits to commercial banks
- Put interest-rate ceilings on other deposits

Securities Act of 1933 and Securities Exchange Act of 1934
- Required that investors receive financial information on securities offered for public sale
- Prohibited misrepresentations and fraud in the sale of securities
- Created the Securities and Exchange Commission (SEC)

Investment Company Act of 1940 and Investment Advisers Act of 1940
- Regulated investment companies, including mutual funds
- Regulated investment advisers

Table 1 Major Financial Legislation in the United States (2 of 5)

Bank Holding Company Act and Douglas Amendment (1956)

- Clarified the status of bank holding companies (BHCs)
- Gave the Federal Reserve regulatory responsibility for BHCs

Depository Institutions Deregulation and Monetary Control Act (DIDMCA) of 1980

- Gave thrift institutions wider latitude in activities
- Approved NOW and sweep accounts nationwide
- Phased out interest-rate ceilings on deposits
- Imposed uniform reserve requirements on depository institutions
- Eliminated usury ceilings on loans
- Increased deposit insurance to $100,000 per account

Depository Institutions Act of 1982 (Garn-St. Germain)

- Gave the FDIC and the Federal Savings and Loan Insurance Corporation (FSLIC) emergency powers to merge banks and thrifts across state lines
- Allowed depository institutions to offer money market deposit accounts (MMDAs)
- Granted thrifts wider latitude in commercial and consumer lending

Table 1 Major Financial Legislation in the United States (3 of 5)

Competitive Equality in Banking Act (CEBA) of 1987

- Provided $10.8 billion to shore up the FSLIC
- Made provisions for regulatory forbearance in depressed areas

Financial Institutions Reform, Recovery, and Enforcement Act (FIRREA) of 1989

- Provided funds to resolve savings and loan (S&L) failures
- Eliminated FSLIC and the Federal Home Loan Bank Board
- Created the Office of Thrift Supervision to regulate thrifts
- Created the Resolution Trust Corporation to resolve insolvent thrifts
- Raised deposit insurance premiums
- Reimposed restrictions on S&L activities

Federal Deposit Insurance Corporation Improvement Act (FDICIA) of 1991

- Recapitalized the FDIC
- Limited brokered deposits and the too-big-to-fail policy

Table 1 Major Financial Legislation in the United States (4 of 5)

Set provisions for prompt corrective action
Instructed the FDIC to establish risk-based premiums
Increased examinations, capital requirements, and reporting requirements
Included the Foreign Bank Supervision Enhancement Act (FBSEA), which strengthened the Fed's authority to supervise foreign banks
Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994
Overturned prohibition of interstate banking
Allowed branching across state lines
Gramm-Leach-Bliley Financial Services Modernization Act of 1999
Repealed Glass-Steagall and removed the separation of banking and securities industries
Sarbanes-Oxley Act of 2002
Created Public Company Accounting Oversight Board (PCAOB)
Prohibited certain conflicts of interest
Required certification by CEO and CFO of financial statements and independence of audit committee

Table 1 Major Financial Legislation in the United States (5 of 5)

Federal Deposit Insurance Reform Act of 2005

- Merged the Bank Insurance Fund and the Savings Association Insurance Fund
- Increased deposit insurance on individual retirement accounts to $250,000 per account

Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010

- Created Consumer Financial Protection Bureau to regulate mortgages and other financial products
- Required routine derivatives to be cleared through central clearing houses and exchanges
- Required annual bank stress tests
- Limits Federal Reserve lending to individual firms
- Authorized government takeovers of financial holding companies
- Created Financial Stability Oversight Council to regulate systemically important financial institutions
- Banned banks from proprietary trading and from owning large percentages of hedge funds